



Birla Institute of Technology & Science, Pilani

Pilani | Dubai | Goa | Hyderabad

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 1 August 2026

In addition to part I (General Handout for all courses, appended to the Timetable), this portion provides further specific details about the course.

Course No.: ECE F331

Course Title: Microwave Engineering

Instructor-in-charge: Prof. Dinesh Rano (dinesh.rano@pilani.bits-pilani.ac.in, 2220-C)

Instructor: Prof. Praveen Kumar A. V. (praveen.kumar@pilani.bits-pilani.ac.in) and Mr. Satyam (p20230042@pilani.bits-pilani.ac.in)

- 1. Course description:** Review of Maxwell's equations; Wave propagation: Normal & Oblique incidence; Transmission Lines & Smith Chart, Waveguide: Rectangular & Cylindrical waveguides; Dielectric Waveguides; Cavity resonator, equivalent Circuit Model, characteristics; Microwave devices (Waveguide based and planar); S parameters; Measurement & Impedance Matching; Microwave Generators & Amplifiers; Antennas and its characteristics and types.
- 2. Scope and Objective:** This course provides a comprehensive understanding of electromagnetic wave theory and microwave engineering, focusing on Maxwell's equations, wave propagation, and transmission line theory. It covers the analysis and design of waveguides, cavity resonators, and dielectric structures. Students will explore microwave devices, S-parameters, impedance matching techniques, and measurement methods. The course also introduces microwave generators, amplifiers, and various types of antennas with their characteristics. The objective is to equip students with the theoretical foundation and practical skills needed to analyze, design, and apply high-frequency systems in modern communication, radar, and sensing applications
- 3. Course learning outcomes:**
 - CLO1:** Model and analyze discrete components and distributed transmission lines using fundamental microwave theory and transmission line equations.
 - CLO2:** Design and simulate microstrip lines and load terminations to evaluate scattering-parameter performance.

CLO3: Apply S-parameter techniques to characterize and assess the performance of passive microwave components.

CLO4: Identify and evaluate various antenna types and configurations, including dipole antennas, antenna arrays, patch antennas, and broadband antennas for specific radiation patterns and applications.

4. Textbooks and references:

TB1. David M. Pozar, Microwave Engineering, 4th edition. Wiley, 2015

TB2. C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Hoboken, NJ: John Wiley & Sons, 2016, ISBN 978-1-118-64206-1

RB1. Microwave Devices and Circuits, Third Edition, SAMUEL Y. LIAO, PRENTICE HALL

5. Course Format & Weekly Workflow

weekly schedule for 14 weeks with [3] lectures per week

Wk no.	Module no.	Description	Ref.	Learning Outcome
1	1	Introduction to radio frequency engineering,	Slides	Various frequency bands, microwave passive components, and high frequency modeling of RF circuits - resistors, capacitors, and inductors, Q-factor, Bandwidth
2	1	Maxwell's equations	TB1 – Ch- 2, Slides	Displacement current analysis, continuity equation, Ampere's law
2	2	Network analysis	TB1 – Ch-4, Slides	Microwave network analysis - Multi-Port Networks, and S-Matrix, Scattering parameters - two, three, and four port analysis
3	2	Network analysis	TB1 – Ch-4, Slides	Matched/lossless/reciprocal Networks and Signal flow graph
4	3	Transmission Lines	TB1 – Ch-3, slides	Microstrip lines, quasi-TEM mode, design aspects, phase and group velocity, dispersion (briefly),
5	3	Transmission Lines	TB1 – Ch-3, slides	Frequency-dependent effects and higher-order modes, design of frequency-dependent loads (FDCLs) with lumped elements
6	4	Smith chart and Impedance Matching	TB1 – Ch-3, Slides	Introduction to the Smith chart, Impedance matching techniques, multi-stage quarter-wave transformer, reflection analysis

7	5	Rectangular Waveguides	TB1 – Ch-6, Slides	TE, TM, and TEM modes: Cut-off frequency, Guide wavelength
8	5	Rectangular Waveguides	TB1 – Ch-6, Slides	Group and phase velocity, Field pattern for TE, TM, and TEM modes
8	6	Microwave components	TB1 – Ch-7, Slides	Power dividers and couplers, basic Properties, PD-design aspects, even and odd mode analysis, Wilkinson Power Divider
9	6	Microwave components	TB1 – Ch-7, Slides	Quadrature coupler, coupled-line coupler, RF filter design, Filter implementation using microstrip lines and lumped components
10	7	Microwave Generators and Amplifiers	Class notes and RB1	To qualitatively and quantitatively analyze low and high-power microwave generators and amplifiers
11	8	Antenna design principles	TB2: Chapter 2/Slides	Introduction – principle, Antenna parameters, Radio communication link, Antenna field zone, Directivity, Gain, Effective aperture
12	8	Antenna design principles contd.	TB2: Chapter 2/Slides	Equivalent circuit model, Efficiency, Polarization, Friis Transmission
13	8	Linear/wire antennas	TB2: Chapter 4/Slides	Linear Wire Antennas – Infinitesimal (Hertz) Dipole - Near field and far-field, Radiation resistance, Directivity, and effective aperture
14	8	Advanced antennas	TB2: Chapter 14/Slides	Microstrip patch antenna, polarization aspects, narrow/broadband antenna

6. Evaluation Scheme:

Component	Duration	Weightage	Marks	Date and time	Nature of Component
Mid-semester Test	90 min	30 %	60	09/10/2026 AN1	Closed Book
Quizzes (4)	During tut hours	15 %	30	During tut hours	Closed Book (to be announced in class)
Lab Evaluation (regularity + lab exam)	TBA	20 %	40	TBA	Open
Comprehensive. Exam.	180 min	35 %	70	14/12/2026 FN	Closed /Open Book
Total			200		

Note:

Quiz evaluation: Two before mid-semester and two before comprehensive. All quizzes are compulsory.

Lab evaluation: Attendance for each lab (2 make-up labs, one before mid-semester + one before final exam) + lab performance (throughout the semester), **mid-semester exam: viva, final lab exam:** lab performance (lottery-based) + viva

7. **Minimum performance requirement for a valid grade:** A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student receives marks below the prescribed standard mentioned above, he/she may be awarded an NC report.
8. **Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
9. **Chamber consultation hours:** Saturday (10 to 11 am), Thursday (6 to 7 pm)
10. **Makeup Policy:** [Makeup will be given only for genuine cases (accompanied by proof) for the Mid-semester or comprehensive exam only. No makeup for lab exam or quizzes.

Dinesh Rano
Instructor-in-charge
ECE F331



FIRST SEMESTER 2026-27

Course No : ECE_F331
Course Title : MICROWAVE ENGINEERING LAB
Instructor-in-Charge : Dinesh Rano
Lab Instructor : Abhishek Karwa (RS), Himanshi Chaudhary (RS), Pradeep Verma (RS), Laveena Dadhich (RS)

List of Experiments

Simulation based on CST

- 1. Design and analysis of microstrip lines (probably 1 – 3 labs):**
 - a. Varying width (W) and substrates (ϵ_r) $\rightarrow$ plot Z_0 vs W for different ϵ_r (at least 10 points)
 - b. Design load terminations: matched/unmatched with those of the microstrip line's characteristic impedance $\rightarrow$ port and discrete impedance
 - c. Design of a quarter-wave transformer matching network for single, double, triple stages, plot Γ vs f
 - d. Design a $\lambda/2$ line and analyze the input impedance
- 2. Design of matching networks: (2 labs)**
 - a. Design of FDCL (frequency-dependent complex loads) at 2.45, 5.8 GHz: Emulation of inductive and capacitive loads with the TX line
 - b. Design of an impedance matching network with series/shunt stub
- 3. Design of planar microwave devices: (2 labs)**
 - a. Waveguides
 - b. E- and H-plane Tees
 - c. Coupler and attenuator (if possible)
- 4. Introduction to antenna design: (2 labs)**
 - a. Design of dipole and monopole patch antennas
 - b. Design different feeding mechanisms
 - c. Introduction to FSS



Hardware based on Waveguide Bench

1. a. **STUDY OF MICROWAVE COMPONENTS: Familiarization with Microwave Lab Equipment**
b. **MODE CHARACTERISTICS OF THE REFLEX KLYSTRON: Characterization of Reflex Klystron modes (5 modes) (2 labs)**
2. **GUNN DIODE CHARACTERISTICS (1 lab): Study the V-I Characteristics of the Gunn Diode and the depth of modulation of the PIN Diode. (1 lab)**
3. **DETERMINATION OF VSWR AND REFLECTION COEFFICIENT (1 lab): To become familiar with the basic technique for measuring Voltage Standing Wave Ratio (VSWR).**
4. **STUDY OF E-, H-PLANE TEE AND MAGIC TEE CHARACTERISTICS (1 lab) AND STUDY OF DIRECTIONAL COUPLER CHARACTERISTICS (1 lab)**
5. **STUDY OF ATTENUATOR AND CIRCULATOR CHARACTERISTICS (1 LAB) AND DETERMINATION OF THE S-MATRIX**
6. **MEASUREMENT OF POWER RECEIVED AND GAIN FOR AN ANTENNA SETUP WITH VNA AND A SPECTRUM ANALYZER: (1 LAB)**
 - a. **Horn antenna**
 - b. **Patch antenna**

Dinesh Rano